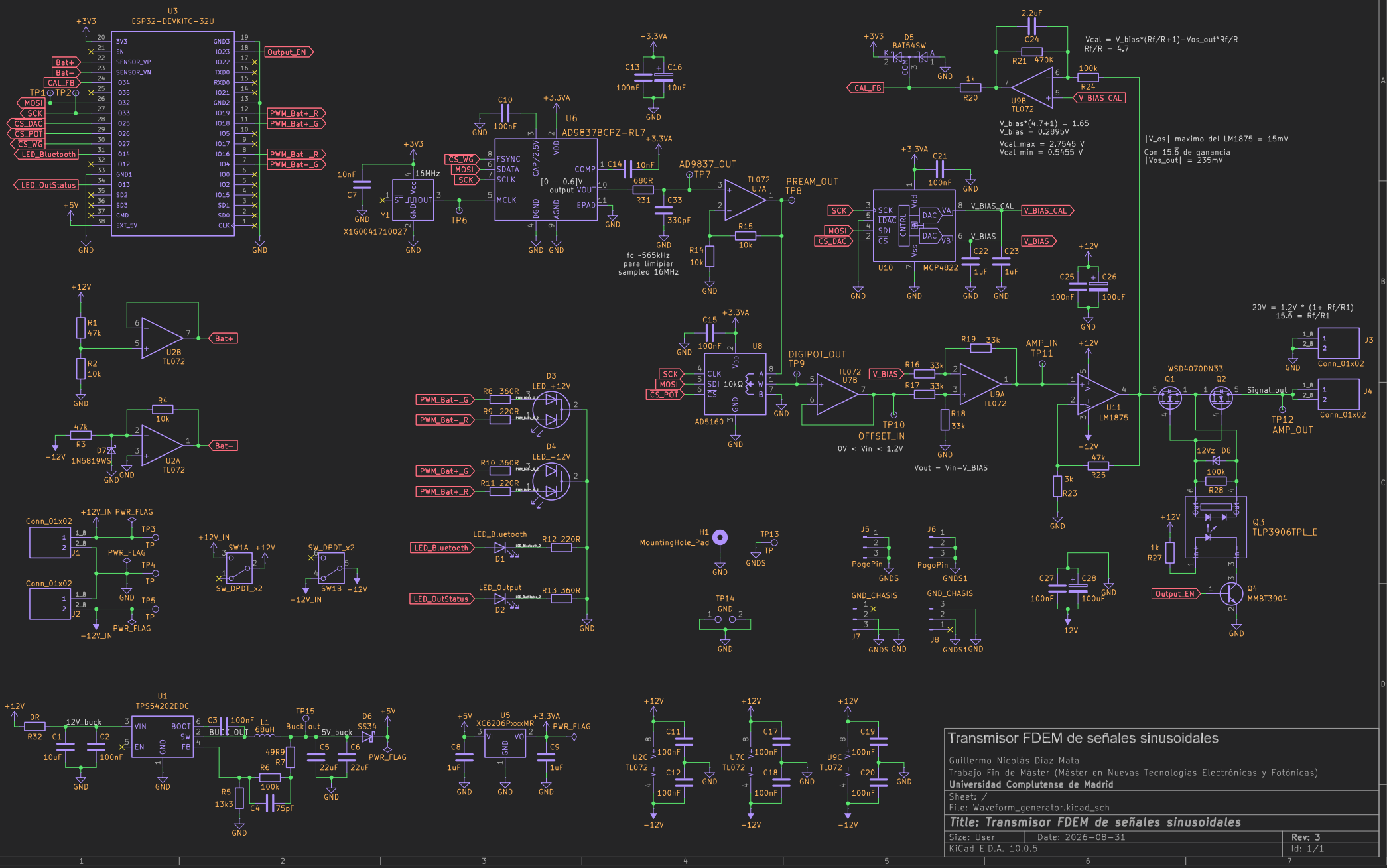

U3
ESP32-DEVKITC-32U
+3V3
Bat+
Bat-
CAL_FB
MOSI
SCK
CS_DAC
CS_POT
CS_WG
LED_Bluetooth
LED_OutStatus
+5V
GND
Output_EN
PWM_Bat+_R
PWM_Bat+_G
PWM_Bat-_R
PWM_Bat-_G
U6
AD9837BCPZ-RL7
FSYNC
SDATA
SCLK
MCLK
COMP
VOUT
EPAD
AD9837_OUT
TP7
PREAM_OUT
TP8
TL072
U7A
R31 680R
C33 330pF
R14 10k
R15 10k
16MHz
X1G0041710027
C7 10nF
C10 100nF
C13 100nF
C16 10uF
C14 10nF
fc ~565kHz
para limpiar
sampleo 16MHz
[0 - 0.6]V
output VOUT
U10
MCP4822
V_BIAS_CAL
V_BIAS
C21 100nF
C22 1uF
C23 1uF
C24 2.2uF
R21 470K
R24 100k
R20 1k
D5
BAT54SW
U9B
TL072
Vcal = V_bias*(Rf/R+1)-Vos_out*Rf/R
Rf/R = 4.7
V_bias*(4.7+1) = 1.65
V_bias = 0.2895V
Vcal_max = 2.7545 V
Vcal_min = 0.5455 V
|V_os| maximo del LM1875 = 15mV
Con 15.6 de ganancia
|Vos_out| = 235mV
20V = 1.2V * (1+ Rf/R1)
15.6 = Rf/R1
C25 100nF
C26 100uF
U2B
TL072
R1 47k
R2 10k
Bat+
U2A
TL072
R4 10k
R3 47k
D7
1N5819WS
Bat-
C15 100nF
U8
AD5160
10kΩ
DIGIPOT_OUT
TP9
TL072
U7B
R16 33k
R17 33k
R19 33k
R18 33k
U9A
TL072
AMP_IN
TP11
OFFSET_IN
TP10
0V < Vin < 1.2V
Vout = Vin-V_BIAS
U11
LM1875
R23 3k
R25 47k
WSD4070DN33
Q1
Q2
Signal_out
TP12
AMP_OUT
J3
J4
Conn_01x02
D3
LED_+12V
D4
LED_-12V
R8 360R
R9 220R
R10 360R
R11 220R
PWM_Bat+_G
PWM_Bat+_R
PWM_Bat-_G
PWM_Bat-_R
LED_Bluetooth
LED_OutStatus
D1
D2
R12 220R
R13 360R
12Vz D8
100k
R28
Q3
TLP3906TPL_E
1k
R27
Q4
MMBT3904
Output_EN
C27 100nF
C28 100uF
H1
MountingHole_Pad
TP13
GNDS
TP14
J5
J6
PogoPin
GNDS1
GND_CHASIS
J7
J8
Conn_01x02
J1
J2
+12V_IN
PWR_FLAG
TP3
TP4
TP5
SW1A
SW1B
SW_DPDT_x2
U1
TPS54202DDC
VIN
EN
BOOT
SW
FB
GND
R32 0R
12V_buck
C1 10uF
C2 100nF
C3 100nF
BUCK_OUT
L1 68uH
TP15
Buck out
5V_buck
D6
SS34
PWR_FLAG
R7 49R9
R6 100k
R5 13k3
C4 75pF
C5 22uF
C6 22uF
U5
XC6206PxxxMR
C8 1uF
C9 1uF
C11 100nF
C12 100nF
C17 100nF
C18 100nF
C19 100nF
C20 100nF
U2C
U7C
U9C
TL072
-12V
Transmisor FDEM de señales sinusoidales
Guillermo Nicolás Díaz Mata
Trabajo Fin de Máster (Máster en Nuevas Tecnologías Electrónicas y Fotónicas)
Universidad Complutense de Madrid
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Date: 2026-08-31
Rev: 3
KiCad E.D.A. 10.0.5
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